ANSWER KEY								AT#01(Kinematics)	
Q.	1	2	3	4	5	6	7	8	9
A.	**3**	**3**	**1**	**4**	**2**	**3**	**1**	**1**	**4**
Q.	10	11	12	13	14	15	16	17	18
A.	**2**	**2**	**2**	**4**	**4**	**2**	**4**	**4**	**2**
Q.	19	20	21	22	23	24	25	26	27
A.	**3**	**4**	**3**	**4**	**4**	**2**	**3**	**3**	**3**
Q.	28	29	30	31	32	33	34	35	36
A.	**1**	**3**	**2**	**4**	**1**	**1**	**3**	**2**	**1**
Q.	37	38	39	40	41	42	43	44	45
A.	**1**	**2**	**2**	**3**	**2**	**4**	**3**	**4**	**3**

Hints And Solution

1. ANS- 3

We know that,

$v = \frac{dx}{dt}$, it is slope of x-t graph from,

0 sec – 5 sec – slope will be decreasing 5 sec – slope is zero

5 sec - 10 sec – slope will be increasing in but a negative value.

2. ANS- 3

Height of tap = 5m and g= 10 m/s^2

For the first drop,

$5 = ut + \frac{1}{2}gt^2 \rightarrow$

$5 = (0 \times t) + \frac{1}{2}\, 10\, t^2 \rightarrow t = 1\, sec$

It means that the third drop leaves after one second of the first drop. Or, each drop leaves after every 0.5 sec.

Distance covered by the second drop in 0.5 sec

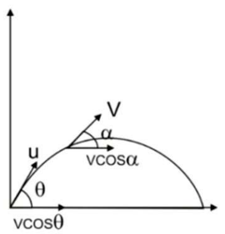
$S = ut + \frac{1}{2}gt^2$

$= (0 \times 0.5) + \frac{1}{2} \times 10 \times (0.5)^2 = 1.25m$

Therefore, distance of the second drop above the ground = 5 – 1.25 = 3.75 m

3. ANS-1

During the projetile motion,



Horizontal component of velocity is always same

$vcos\alpha = ucos\theta$

Or, $v = \frac{ucos\theta}{cos\alpha}$

Or, $ucos\theta sec\alpha$

4. ANS- 4

Given x = 0.20m, y =0.20m, u=1.8m/s

Let the ball strike the nth step of stairs, vertical distance travelled = ny, Using equation of motion in y direction

S = ut + 1/2at^2

$$\Rightarrow \quad -ny = 0 - \frac{1}{2}gt^2 \text{..........(1)}$$

Horizontal distance travelled = nx

$$\Rightarrow \quad nx = ut$$

$$\Rightarrow \quad t = \frac{nx}{u} \text{..........(2)}$$

Using equation (1) and (2)

$$ny = \frac{1}{2}g\left(\frac{nx}{u}\right)^2$$

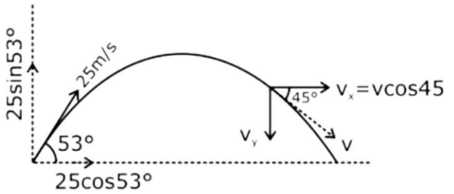
$$or \quad nx = \frac{2u^2}{g}\frac{y}{x^2} = 3.3 \Rightarrow 4^{th} \ stair$$

5. ANS- 2

At E, the slope of the curve is negative i.e. for a displacement-time graph the slope represents the velocity and at E, the velocity is negative as the slope is negative.

6. ANS- 3

Horizontal component of velocity, throughout the motion remains constant



using $u\cos\theta = v\cos\alpha$

$$25\cos 53^0 = v\cos 45^0$$

$$\Rightarrow v = 25 \times \frac{3}{5}\sqrt{2} = 15\sqrt{2}$$

$$v_y = v\sin 45^0 = 15$$

Now using $v_y = u_y + a_y t$ in y direction

$$-15 = 25\sin 53^0 - gt$$

$$-15 = 25 \times \frac{4}{5} - 10t \quad \Rightarrow t = 3.5s$$

7. ANS- 1

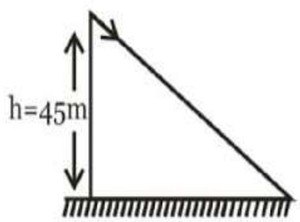
$$From\ the\ given\ curve, v = -\frac{v_0}{x_0}x + v_0$$

$$Because\ , a = v\frac{dv}{dx}x = \left(-\frac{v_0}{x_0}x + v_0\right)\left(-\frac{v_0}{x_0}\right)$$

$$= \left(\frac{v_0}{x_0}\right)^2 x - \frac{{v_0}^2}{x_0}$$

8. ANS- 1

Taking motion in vertical direction - u = 0, g = 10m/s^2 , h= 45 m



h= ut + $\frac{1}{2}$ gt^2

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2(45)}{10}} = 3sec$$

(b) taking motion in horizontal direction

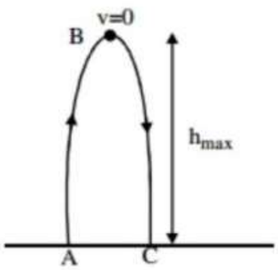
u=0 , a= 10m/s^2, t = 3 sec

x= ut + $\frac{1}{2}$ at^2 = 45m

9. ANS- 4

The ball traverse distance upward and then downwards.

Hence, time taken is going from A to B is $\frac{t}{2} = \frac{4}{2} = 2s$



Also at the maximum height B,speed is zero.

Therefore

v = u – g t_{max}

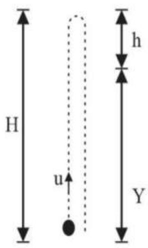
Where v=0 , t_{max}= 2s, g = 10m/s^2 ∴ u = g

t_{max} = 10 ×2 =20 m/s

10. ANS -2

Let total height=H, Total Time of ascent=T

So, H= uT- $\frac{1}{2}$gT^2



Distance covered by ball in time(T-t)sec is

y= u(T-t)-$\frac{1}{2}$g$(T-t)^2$

So, distance covered by ball in last t sec

h=H-y= [uT-$\frac{1}{2}$gT^2]-[u(T-t)-$\frac{1}{2}$g$(T-t)^2$]

By solving and putting T=$\frac{u}{g}$

We will get,

h=$\frac{1}{2}$gt^2

11. ANS- 2

Using equation of trajectory

$y = x\tan\alpha\left(1 - \frac{x}{R}\right)$

For point (3,4)

$4 = 3\tan\alpha\left(1 - \frac{3}{R}\right)$ ....(1)

For point (4,3)

$3 = 4\tan\alpha\left(1 - \frac{4}{R}\right)$ ....(2)

$dividing\ (1) and\ (2)$and solving we get R = $\frac{37}{7}m$

12. ANS – 2

$h = \frac{1}{2}gT^2$ or $T = \sqrt{\frac{2h}{g}}$

Now S = $\frac{1}{2}g\left(\frac{T}{3}\right)^2 = \frac{g}{18} \times \frac{2h}{g} = \frac{h}{9}$

Height above the ground= $h - \frac{h}{9} = \frac{8h}{9}$

13. ANS-4

Since |Displacement|≤ Distance

$\therefore \frac{|Displacement|}{time} \leq \frac{Distance}{time}$

|Average velocity|≤ Average speed

14. ANS-4

x=a$e^{-\alpha t}+be^{\beta t}$

$\frac{dx}{dt}$= $-$a$\alpha e^{-\alpha t}+b\beta e^{\beta t}$

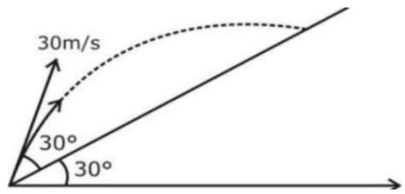
V= $b\beta e^{\beta t}$$-a\alpha e^{-\alpha t}$

As t increases, $e^{-\alpha t}$ decreases and $e^{\beta t}$ increases. So v will increase with time

15. ANS-2

Range on incline plane,

R=2u2sin(α-β)cosα / gcos2β



Here, =30°and α=30°+30°=60°

R=2(30)2sin(30°)cos(60°) / 10cos^2(30°)

$R=\frac{2(30)2(\frac{1}{2})(\frac{1}{2})}{10(\frac{3}{4})}$

R =60m

16. ANS- 4

Range of projectile R= $\frac{u^2 sin2\theta}{g}$

for θ = 60°, R = $\frac{20^2 sin120°}{g}$ = $20\sqrt{3}$ = 34.64 m

for % error $\frac{\Delta R}{R}$ = $\frac{2\Delta u}{u}$

ΔR= $\frac{2 \times 5}{100}$x $20\sqrt{3}$ = $2\sqrt{3}$ = 3.464 m

So, range R = 34.64 ± 3.46 m

R_{min}= 31.1 m and R_{max} = 38.1 m

17. ANS- 4

Let $\overrightarrow{u_1}$ and $\overrightarrow{u_2}$ be the initial velocities of the two particles and θ_1 and $\theta_2$be their angles of pprojection with the horizontal

The velocities of the two particles after time t are,

$\vec{v}_1=(u_1\cos\ \theta_1)\hat{i}+(u_1\sin\ \theta_1-gt)\hat{j}$and

$\vec{v}_1=(u_1\cos\ \theta_1)\hat{i}+(u_2\sin\ \theta_2-gt)\hat{j}$

Their relative velocity is

$\vec{v}_{12}=\vec{v}_1-\vec{v}_2$

$=(u_1\cos\ \theta_1-u_2\cos\ \theta_1)\hat{i}++(u_1\sin\ \theta_1-u_2\sin\ \theta_2)\hat{j}$

Which is a constant. So the path followed by one, as seen by the other straight line, making a constant angle with the horizontal.

18. ANS-2

Since acceleration is constant, therefore there is uniform increase in velocity. So, the v-t graph is straight line slopping upward to the right. When acceleration becomes zero, velocity is constant. So v-t graph is a straight line parallel to the time axis

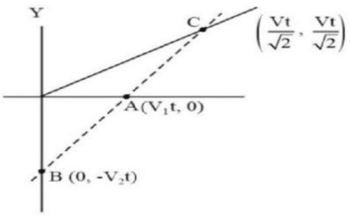
19. ANS- 3

Area under graph will give displacement height

$$h=\frac{1}{2}\times 2\times 3.6+3.6\ \times 8\ +\frac{1}{2}\times 2\times 3.6=36\,m$$

20. ANS-4

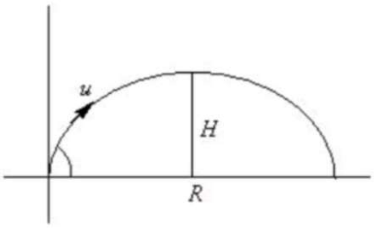
Slope of line AC = slope of line BA



$$\frac{(\frac{vt}{\sqrt{2}} - 0)}{(\frac{vt}{\sqrt{2}} - v_1 t)} = \frac{0 - (-v_2 t)}{(v_1 t - 0)}$$

$$v = \frac{\sqrt{2} v_1 v_2}{v_2 - v_2}$$

21. ANS- 3



The range of the particle is

$R = \frac{u^2 \sin(2\theta)}{g}$, $at = 45°$; $R = \frac{u^2}{g}$ .... (i)

$H = \frac{u^2 \sin^2(\theta)}{2g}$, $at = 45°$; $H = \frac{u^2}{4g}$ .... (ii)

Dividing eqs. (i) byeqs.(ii)

$$R = 4H$$

22. ANS- 4

V_r =velocity of rain

V_{rm} =velocity of rain with respect to man

V_m =velocity of man

$For\ first\ case\ rain\ appear\ vertical,$

$Hence - V_{rx} = 5\ \text{km/h}$

$For\ second\ case - tan60^0 = \frac{V_{ry}}{V_{rx}}$

$\Rightarrow V_{ry} = 5\sqrt{3}\ \text{km/h}$

$The\ velocity\ of\ rain\ with\ respect\ to\ running\ man$

$$= V_{ry} = 5\sqrt{3}\ \text{km/h}$$

23. ANS- 4

Given initial velocity = v_0

Final velocity =0

$a = -x^2$

Now we know that $a = \frac{dv}{dt} = \frac{dv}{dt} \times \frac{dx}{dx} = v\frac{dv}{dx} = -x^2$

$vdv = -x^2 dx$

Integrating we get

$$\int_{v_0}^{0} vdv = \int_0^x -x^2 dx$$

$$\left(\frac{v^2}{2}\right)_{v_0}^{0} = -\left[\frac{x^3}{3}\right]_0^x$$

$$-\frac{v_0^2}{2} = -\frac{x^3}{3}$$

$$x = \left(\frac{3v_0^2}{2}\right)^{\frac{1}{3}}$$

24. ANS- 2

Horizontal velocity remains same
30cos60° =v cos45°
V = $15\sqrt{2}\ m/s$

25. ANS-3

Let after t time two bodies are 40 m apart.

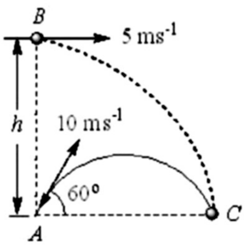
Then according to the problem

$$\frac{1}{2}gt^2-\frac{1}{2}g(t-2)^2=40$$

$\Rightarrow 2t-2=4 \text{ or } t=3s$

26. ANS-3

For both the particles to collide at point, their time of flight should be same



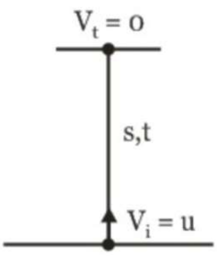
$$T=\frac{2u\sin\theta}{g}=\frac{2\times 10\times \sin 60}{10}=\sqrt{3}s$$

Time of flight of particle thrown from above

$$T=\sqrt{\frac{2H}{g}}=\sqrt{3}s \Rightarrow H=15m$$

27. ANS- 3

Total displacement is 0 as the particle returns to the original position, thus the average velocity is zero. Total distance travelled is 2S and total time taken is 2t



$0^2=u^2-2gs$

$\therefore s=\frac{u^2}{2g}$

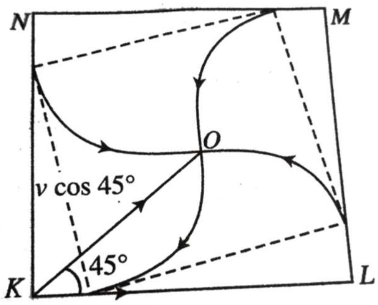
also $0=u-gt$

$\therefore t=u/g$

$\therefore 2t=2u/g$

Speed = $\frac{u^2}{g}/\frac{2u}{g}=\frac{u}{2}$

28. ANS- 1



The velocity of *K* throughout the motion towards the center of square is $v\cos 45°$ and the displacement covered by this velocity will be *KO*

$$t=\frac{KO}{v\cos 45°}=\frac{\left(\sqrt{2}\frac{d}{2}\right)}{\frac{v}{\sqrt{2}}}=\frac{d}{v}$$

29. ANS- 3

30. ANS- 2

31. ANS - 4

Angle made by projectile with horizontal is 30° which is less than 45°. Hence the two velocities are not perpendicular at any instant.

32. ANS - 1

Since the time of flight depends upon the vertical component of projection velocity, rotation of earth doesn't affect the time of flight.

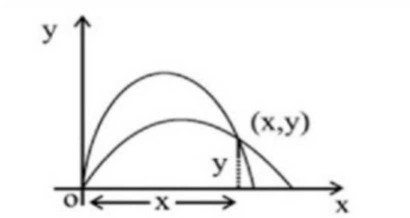
33. ANS – 1

The time for the particle 1 to pass through common point of their path

$$t_1=\frac{x}{u_1}$$

The time for particle 2 to pass through common point of their path

$$t_2 = \frac{x}{u_2}$$



Difference in time $t = t_1 - t_2$

Time $t = \frac{x}{u_1} + \frac{x}{u_2} = x\left(\frac{1}{u_1} - \frac{1}{u_2}\right)$

$$y = x\tan\theta - \frac{gx^2}{2u_x^2} \quad y = x\frac{v_1}{u_1} - \frac{gx^2}{2u_1^2} = x\frac{v_2}{u_2} - \frac{gx^2}{2u_2^2}$$

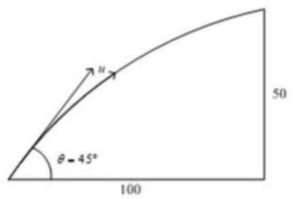
$$\frac{v_1}{u_1} - \frac{v_2}{u_2} = x\frac{g}{2}\left(\frac{1}{u_1^2} - \frac{1}{u_2^2}\right)$$

$$\frac{v_1}{u_1} - \frac{v_2}{u_2} = x\frac{g}{2}\left(\frac{1}{u_1} + \frac{1}{u_2}\right)\left(\frac{1}{u_1} - \frac{1}{u_2}\right)$$

$$t = x\left(\frac{1}{u_1} - \frac{1}{u_2}\right) = \frac{2}{g}\left(\frac{v_1u_1 - v_2u_2}{u_1 + u_2}\right)$$

34. ANS- 3

From the equation of trajectory –



$$y = x\tan\theta - \frac{gx^2}{2y^2cos^2\theta}$$

$$50 = 100\tan 45º - \frac{10(100)^2}{2y^2cos^2 45º}$$

$$u^2 = \frac{100^2}{5}$$

$u = 20\sqrt{5}$m/s

35. ANS- 2

Let u be the initial upward velocity of the ball from A (the top of the tower) and let h be the height of the tower. Taking the downward motion of the first stone from A to the ground is positive, we have

$$h = -ut_1 + \frac{1}{2}gt_1^2 \ldots(i)$$

Taking the downward motion of the second stone from A to the ground is positive, we have

$$h = ut_2 + \frac{1}{2}gt_2^2 \ldots(ii)$$

Multiplying eq (i) by t_2 and eq (ii) by t_1 and adding we get

$$h(t_1 + t_2) = \frac{1}{2}gt_1t_2(t_1 + t_2)$$

$$so, h = \frac{1}{2}gt_1t_2 \ldots(iii)$$

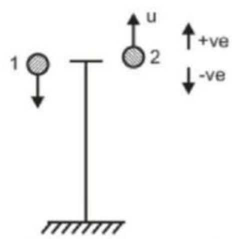
For stone falling under gravity from the top of the tower

$$h = \frac{1}{2}gt_3^2 \ldots(iv)$$

fromeq (iii) and (iv),

$$t_3 = \sqrt{t_1t_2} = \sqrt{9 \times 4} = 6\,s$$

36. ANS- 1



$$s_1(t) = \frac{1}{2}gt^2(downwards)and$$

$$s_2 = ut - \frac{1}{2}gt^2\ (upwards)$$

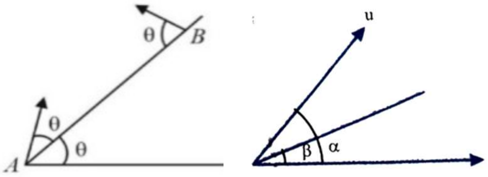
Distance between the two stones will be

$s = s_1(t) + s_2(t) = ut$

Therefore, s –t graph will be a straight line passing through origin.

37. ANS- 1

Here, α = 2θ , β = 0



Time of flight of $T_1 = \frac{2u\sin(\alpha-\beta)}{g\cos\beta}$

$= \frac{2u\sin(2\theta-\theta)}{g\cos\theta} = \frac{2u}{g}\tan\theta$

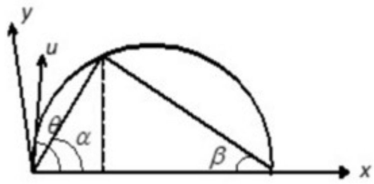
Time of flight of $T_2 = \frac{2usi\ \ \theta}{g\cos\theta} = \frac{2u}{g}\tan\theta$

So, $T_1 = T_2$. The acceleration of both the particles is g downwards. Therefore, relative acceleration between the two is zero or relative motion between the two is uniform. The relative velocity of A w.r.t. B is towards AB, therefore collision will take place between the two in mid air.

38. ANS- 2

If velocity is constant then both magnitude i.e. speed and direction remains constant and acceleration is zero.But if speed is constant then direction of velocity may change and therefore it may have acceleration.

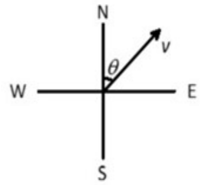
39. ANS- 2



$\tan\theta = \tan\alpha + \tan\beta$

$\tan\theta = \tan 30° + \tan 60° = \frac{4}{\sqrt{3}}$

40. ANS - 3



$u = v_y$ at $v\cos 45°$ $t = \frac{v}{a\sqrt{2}}$

when v_{man} due north is equal to wind velocity component along *north,*

then wind is found to blow due east.

$v_m = v\cos 45°$

$v_m = \frac{v}{\sqrt{2}}$

$v_m = at$

$\frac{v}{\sqrt{2}} = at$

$\frac{v}{\sqrt{2}a} = t$

41. ANS- 2

In one dimentional motion , the body can have at a time one velocity but not two values of velocities.

42. ANS - 4

As the horizontal component is the same for a projectile, Let projectile velocity is v when particle makes angle 45° with horizontal, Hence,

$$v = u\cos\theta\sec\phi = 20\cos 60^\circ\sec 45^\circ$$

$$v = 20\cdot\frac{1}{2}\cdot\sqrt{2} = 10\sqrt{2}\mathrm{m/s}$$

$$t = \frac{v_y - u_y}{g} = \frac{10\sqrt{3}-10}{10} = (\sqrt{3}-1)s$$

43. ANS- 3

At t = 1 s, the particle is at maximum height. At maximum height velocity is horizontal and acceleration is vertical. Hence, the angle between the velocity vector and acceleration vector is 90°. Therefore at this position tangential acceleration, i.e., the rate of change of speed with respect to time is zero.

44. ANS-4

Let v be the velocity of projectile at this instant. Horizontal component of velocity remains unchanged. Therefore

$$v\cos 30^\circ = 10\cos 60^\circ$$

$$v\frac{\sqrt{3}}{2} = \frac{10}{2}$$

$$\therefore v = \frac{10}{\sqrt{3}}\mathrm{ms}^{-1}$$

45. ANS-3

The direction of velocity is always tangential to the path. Hence, at the top of the trajectory, velocity is in the horizontal direction.